

IDEA-ONE: National Hackathon for One Health

Title : Antimicrobial Resistance (AMR) Awareness :



“We built an AMR awareness chatbot that educates people and farmers on responsible antibiotic use.”

1. What Are Antibiotics and Antimicrobial Resistance?

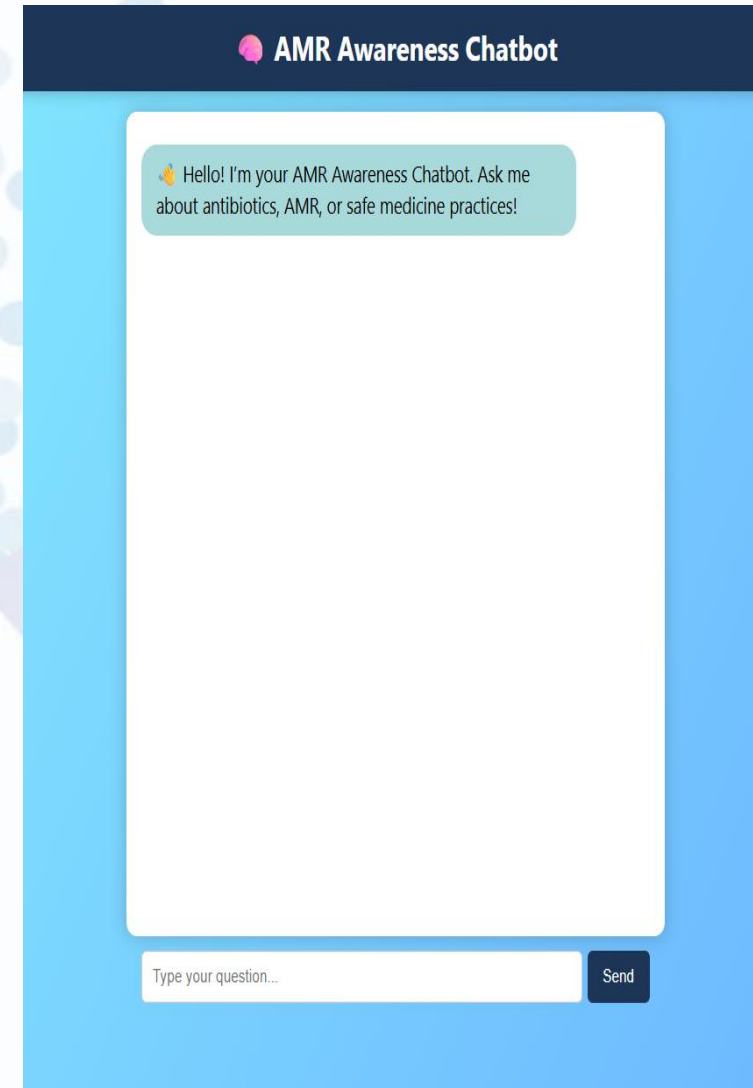
- Antibiotics are drugs that kill bacteria — not viruses.
- Used to treat bacterial infections like pneumonia, not cold or flu.
- Misuse of antibiotics causes bacteria to adapt and resist
- AMR = When bacteria no longer respond to antibiotics.
- It makes infections harder to cure and increases medical costs.

Problem Statement :

- **Creating a chat bot for to get all information for all this antimicrobial Resistance ??**
- Antimicrobial Resistance (AMR) is rapidly increasing due to the overuse and misuse of antibiotics in humans and animals.
- In animal husbandry, antibiotics are often used for faster growth and disease prevention without medical supervision.

Our Solution

- We developed an AMR Awareness Chatbot to educate citizens and farmers about the safe and rational use of antibiotics.
- The chatbot answers common health and antibiotic-related questions in a simple, interactive way.
- It helps users understand what AMR is, how it spreads, and how to prevent it.
- In future, it can be extended to track antibiotic usage in animal farms and alert for overuse patterns.



⚙️ How It Works

1. User Interaction

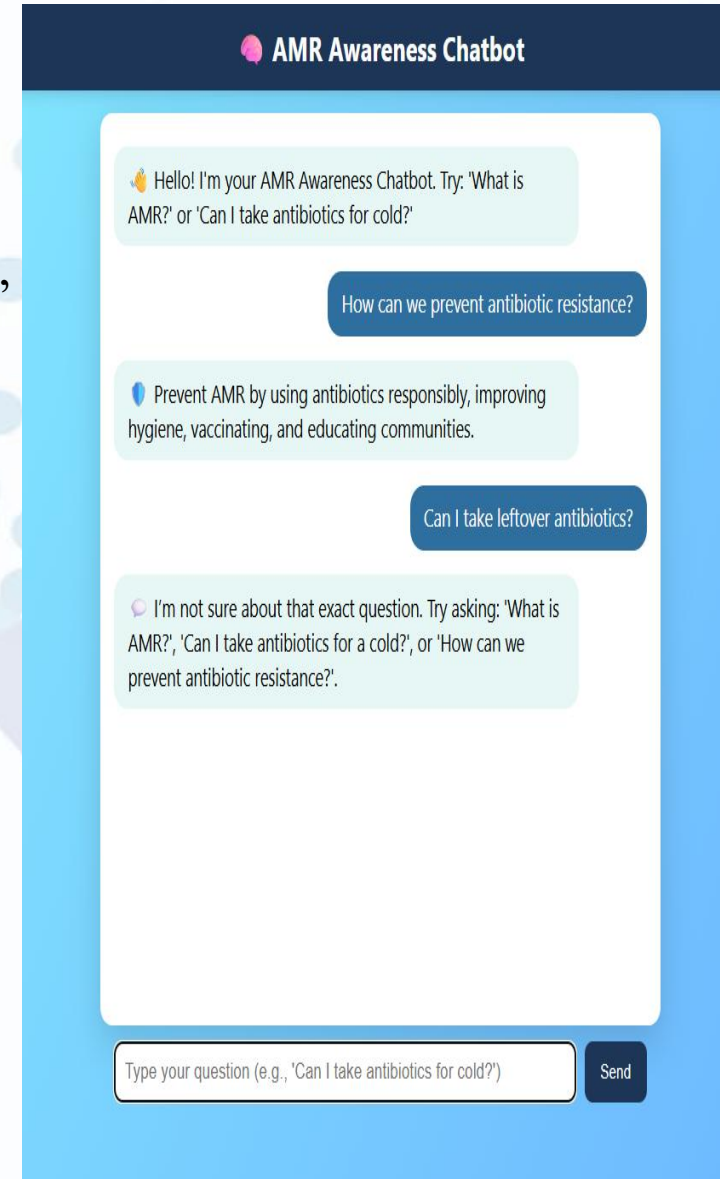
- Citizens or farmers chat with the AMR Awareness Bot on Telegram or web.
- They can ask questions like “Can I take antibiotics for fever?” or “What is AMR?”

2. Smart Response System

- The bot provides instant, accurate answers about antibiotic use, AMR prevention, and awareness.
- Replies are designed in simple, understandable language.

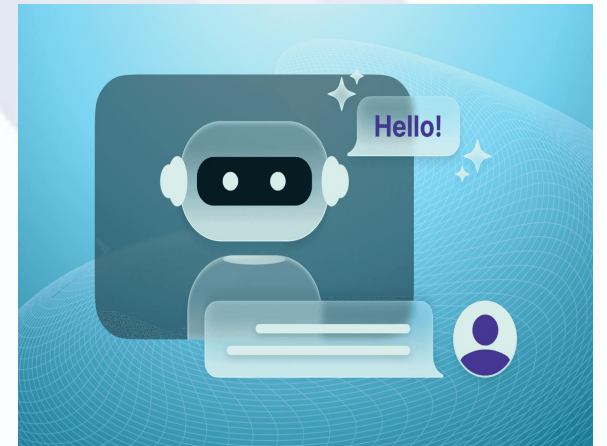
3. Awareness & Education

- Encourages responsible medicine practices.
- Reduces misuse in both humans and livestock through information sharing.



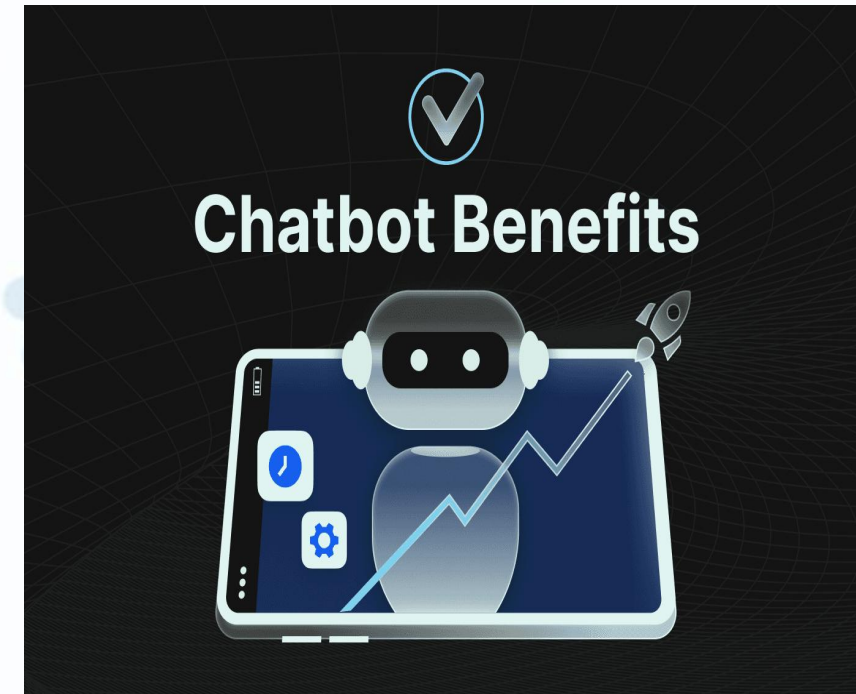
Key Features of the AMR Awareness Chatbot

- Interactive Learning — Users can chat naturally and get answers about AMR, antibiotics, and prevention.
- Simple & User-Friendly Interface — Works on Telegram and web, accessible to all age groups.
- Educational Focus — Promotes safe antibiotic use among citizens, students, and farmers.
- Quick Responses — Provides instant guidance for common doubts like fever, cold, and minor injuries.
- Scalable Design — Can later integrate real-time data from farms to monitor antibiotic use.



🌐 Impact / Benefits of Our Solution :

- Increases Public Awareness — Educates people about responsible antibiotic use and AMR prevention.
- Supports Farmers — Helps them understand the risks of antibiotic overuse in livestock.
- Promotes Rational Drug Use — Reduces self-medication and misuse of antibiotics.
- Encourages Preventive Health Practices — Inspires users to maintain hygiene and consult doctors before taking medicines.
- Scalable for Larger Use — Can be linked with health databases or farm monitoring systems for broader AMR tracking.



OVER VIEW :

Slide No.	Title	Content Summary	Speaker Notes (what to say)
1	Title Slide	“AMR Awareness Chatbot”Team MembersInstitution NameHackathon/Event Name	“Good morning everyone! We’re here to present our project – the AMR Awareness Chatbot, designed to promote responsible antibiotic use and fight antimicrobial resistance.”
2	Problem Statement	Overuse & misuse of antibiotics → AMRPeople unaware of safe practicesLack of simple awareness tools	“Antimicrobial resistance or AMR is a major global threat. People often take antibiotics unnecessarily — for fever, cold, or injuries — leading to resistant bacteria.”
3	What is AMR?	Define AMR & how it spreadsCauses: misuse in humans & animals	“AMR means microorganisms no longer respond to antibiotics. This happens when antibiotics are misused or overused in humans, animals, and farming.”
4	Our Solution: The AMR Awareness Chatbot	Telegram-based chatbotInstant, interactive awareness toolEducates about AMR	“To address this issue, we created a Telegram chatbot that spreads awareness about AMR. It gives clear, quick answers to common questions people have about antibiotics.”
5	How It Works	Built using Python + Telegram APIHandles FAQs on AMRSimple, accessible from any phone	“Our bot is easy to use. Anyone can chat with it on Telegram, ask questions like ‘What is AMR?’ or ‘Can I take antibiotics for cold?’ and get reliable answers instantly.”
6	Sample Questions & Answers	What is AMR?• Can I take leftover antibiotics?• Why should we not overuse antibiotics?• How can farmers reduce AMR risk?	“Here are a few sample questions the chatbot can answer — from everyday health doubts to animal husbandry practices — all promoting responsible antibiotic use.”
7	Technologies Used	• Python• Telegram Bot API• Firebase (optional for data logging)• Gemini API (future integration)	“We used Python and the Telegram Bot API. In future versions, we plan to integrate Gemini or Firebase to make responses more intelligent and track awareness impact.”
8	Impact & Benefits	• Easy awareness tool• Available 24x7• Supports both citizens & farmers• Encourages responsible antibiotic use	“Our chatbot helps people become more aware of AMR anytime, anywhere — reducing the misuse of antibiotics and protecting public health.”
9	Demo Snapshot	Screenshot of your bot chat (e.g., “hi”, “what is AMR?”)Show conversation flow	“Here’s a quick snapshot of our chatbot in action. It’s simple, interactive, and informative — perfect for awareness campaigns.”
10	Future Scope & Conclusion	Future scope:• Regional language support• AI-based responses• School/health awareness integrationConclusion: Small step, big impact	“In conclusion, our chatbot is a small but powerful initiative towards a world free from antibiotic misuse. Thank you!”

Result :

 **AMR Awareness Chatbot (better matching)**

 Hello! I'm your AMR Awareness Chatbot. Try: 'What is AMR?' or 'Can I take antibiotics for cold?'

WHAT IS AMR

 AMR (Antimicrobial Resistance) is when microbes (bacteria, viruses, fungi) no longer respond to medicines made to kill them, making infections harder to treat.

Can i take antibiotics for cold

 No — most colds are viral. Antibiotics are ineffective for viruses and their misuse increases resistance.

Type your question (e.g., 'Can I take antibiotics for cold?')

Send

Tip: type freely — the bot understands many phrasing variations.

THANK
YOU
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